

Name: I Forgot Again Team **Date:** 4/5/24 **Class:** Engineering

MyDesign Portfolio

Element A Initial Template

Element A: Presentation and Justification of the Problem

Problem Statement and Details (A1, A2, A5)

The issue of items falling into the gaps of car seats is a common and frustrating problem many vehicle owners face. This problem involves virtually anyone who drives or rides in a car, as it is an experience shared across demographics and geographies. The cause of the problem is the design of most car interiors, where space exists between the car seat and the center console or door, often referred to as the "car seat gap." This gap can vary but is typically large enough for small to medium-sized items to slip through.

Unique situations where this problem occurs can include moments when items are placed on laps, in pockets, or held in hands, and due to various reasons, such as sudden stops, sharp turns, or even just the act of sitting down or getting up, these items can easily slip into the gap. The size of the problem is considerable, as it affects millions of car owners and passengers worldwide, leading to lost property, potential safety hazards, and unnecessary stress.

The problem physically happens within the interior of a vehicle, specifically between the car seat and the center console or door. It can occur at any time during the use of the vehicle, whether the car is in motion or stationary. The problem happens due to a combination of factors including the design of car seats, the behavior of passengers (such as placing items on their lap), and the physics of objects in motion.

As for how often the problem happens, it's difficult to quantify with precise statistics, but anecdotal evidence suggests that it's frequent for many drivers and passengers. In terms of solutions, various products have been developed to address this issue, such as car seat gap fillers made from materials like silicone, foam, and synthetic leather, designed to fit snugly into the space and prevent items from falling through ([Car Roar: 2023](#); [The Drive: 2020](#); [Car And Truck: 2024](#)).

These solutions prevent the loss of items and contribute to a safer driving experience by minimizing distractions and the need for drivers to reach for dropped items, which can take their attention away from the road ([Car And Truck: 2024](#)). Moreover, such gap fillers can

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help maintain a cleaner and more organized car interior, reducing the time and effort spent retrieving items from difficult-to-reach places ([Car Roar: 2023](#)).

Stakeholder Concerns and Connections (A3, A4)

Addressing the issue of items falling into car seat gaps, the primary stakeholders or stakeholder groups most directly impacted include:

Vehicle Owners and Lessees: They often experience the inconvenience of losing personal items and the potential costs associated with replacing lost items or retrieving them.

Automobile Manufacturers: They are responsible for the design of car interiors and may face consumer dissatisfaction or demands for design changes.

Car Accessory Manufacturers: Companies that produce car seat gap fillers and other related accessories are directly interested in this problem's prevalence.

Parents and Caregivers: They need to ensure the safety and cleanliness of the car environment for children, and items falling into gaps can lead to unhygienic conditions.

Professional Drivers: This group, including taxi, rideshare, and delivery drivers, spends a significant amount of time in their vehicles and may lose valuable work time retrieving items or face distractions while driving.

The concerns of these stakeholders justify the need to address the problem. The frustration and cost of lost items can be significant for vehicle owners and lessees. Automobile manufacturers should consider redesigning car interiors to prevent such issues, which could lead to increased customer satisfaction and potentially higher sales. Car accessory manufacturers see an opportunity to develop products that can solve this problem, leading to increased revenue. Parents and caregivers are concerned about the safety and cleanliness of their vehicles, especially with young children who might put found objects in their mouths. Professional drivers must focus on the road. Distractions caused by dropped items can compromise their safety and efficiency.

The breadth of the problem also has social and environmental impacts on secondary stakeholders. For example, lost items can lead to increased waste if they are not retrieved and are instead replaced. This contributes to the consumption of resources and the generation of waste, impacting the environment. Additionally, the distraction caused by dropped items can lead to road accidents, affecting not only the drivers and passengers but also other road users and emergency services. Developing solutions like car seat gap

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fillers can have positive social impacts by reducing these risks and improving the overall driving experience for everyone involved.

Problem Importance (A6)

In conclusion, the problem of items dropping into car seat gaps is a widespread issue that affects drivers and passengers globally. While it may seem trivial, it can lead to safety concerns and inconvenience. The development of car seat gap fillers represents a practical solution to this everyday problem, enhancing the functionality and safety of car interiors.

Roar, Car. "Do Car Seat Gap Fillers/Organizers Really Work?" Car Roar, 25 Mar. 2022, carroar.com/car-seat-gap-filler/.

Thomas, Andra DelMonico, Lindsay. "Best Car Seat Gap Fillers: Stop Losing Items in Your Vehicle." The Drive, 21 Feb. 2020, www.thedrive.com/reviews/32278/best-car-seat-gap-filler.

"The 10 Best Car Seat Gap Fillers for a Neat and Safe Interior | Car and Truck." Carandtruck.org, 14 Apr. 2024, carandtruck.org/best-car-seat-gap-filler/.